Adrenal & Retroperitoneum

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 145	Surgical and Radiographic Anatomy of the Adrenals
Ch 146	Pathophysiology, Evaluation and Medical Management of Adrenal Disorders
Ch 147	Surgery of the Adrenal Glands
Ch 140	Retroperitoneal Tumours
★	Putting it together: adrenal and retroperitoneal decisions

145 Surgical and Radiographic Anatomy of the Adrenals

P1 *Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma — relations, arteries, veins, zones, embryology*

Not in the paper answer:

- Neonatal gland weight and adrenal rests (§1).

1 · EMBRYOLOGY

- The cortex and gonad share the adrenogonadal ridge, which explains adrenal rests; neural crest forms the medulla.
- At birth the gland is **2×** adult weight. The **fetal zone involutes in year 1**.
- In **renal agenesis** the adrenal is **discoid**.
- **Adrenal rests** occur in ≈1% of adults and in 7.5–15% of neonates along gonadal descent; in congenital adrenal hyperplasia (CAH) they form **testicular masses**.

2 · ZONES AND HORMONES

Cortex is ≈90% and medulla ≈10%. †Outside inwards: **GFR = salt, sugar, sex**.

- Glomerulosa ~15% makes aldosterone; fasciculata **~80%** makes cortisol (lipid-laden "clear cells"); reticularis 5–7% makes DHEAS; the medulla (~10%) holds chromaffin cells.
- The medulla is a **modified sympathetic ganglion** of axonless postganglionic neurones.

3 · RELATIONS AND BLOOD SUPPLY

- Each gland weighs ≈5 g. The right is **retrocaval**, directly above the upper pole; the left lies medial to the upper pole.
- **Exposure**: on the right, liver above, inferior vena cava (IVC) medial, duodenum and kidney below. On the left, pancreas and spleen above, splenic flexure medial.
- Three arteries enter via many twigs, but **one vein drains**. Cortical blood drains into the medulla, so cortisol drives **epinephrine synthesis**.
- The **right vein** is short and enters the **posterior IVC**, so it can avulse. The **left** enters the left renal vein, joined by the inferior phrenic.

> **KEY POINT**
>
> The right side is riskier: high, retrocaval, with a short vein whose avulsion tears the cava.

5 · IMAGING APPEARANCE

- **CT** shows an inverted **V/Y** with limbs of 3–6 mm. Unenhanced **≤10 HU** means lipid-rich.
- **MRI** has the best contrast; on **T2 the gland resembles fat**. Ultrasound separates solid from cystic.
- Adrenal venous sampling (AVS) is the only **functional** study; cavography shows intraluminal tumour.

146 Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

1 · APPLIED ANATOMY AND PHYSIOLOGY

P1 *Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma — veins, arteries, zones, PNMT, COMT forms metanephrines*

Not in the paper answer:

- Anomalous venous drainage occurs in up to **9%**, which matters for adrenal vein sampling (AVS).
- **Pituitary failure or exogenous steroids spare aldosterone**, because the glomerulosa obeys angiotensin II and potassium; the inner zones atrophy without ACTH.

2 · EVALUATION OF AN ADRENAL MASS – PRINCIPLES

KEY POINT

Two questions: **(1) Is it malignant?** (imaging). **(2) Is it hormonally active?** (biochemistry). Yes to either means treat, usually by resection. Imaging cannot identify functional adenomas, so biochemistry is never omitted.

Category	Indications for adrenalectomy (Box 146.1)
Functional	Autonomous cortisol secretion; phaeochromocytoma; lateralised aldosterone excess
Oncological	>4 cm (except myelolipoma); suspicious imaging (lipid-poor, heterogeneous, irregular, infiltrating); growth >1 cm on follow-up; isolated metastasis (multidisciplinary)
Myelolipoma	Very large and/or symptomatic
At nephrectomy for renal cell carcinoma (RCC)	Adrenal abnormal or not visualised; thrombus to adrenal vein level
Bilateral	Failed pituitary surgery for Cushing disease; selected ectopic ACTH; AIMAH; PPNAD (§7)

3 · IMAGING

P2 P4 *Algorithm for management of an incidentaloma; Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — APW/RPW formulas, size–malignancy risk, ESE–ENSAT follow-up*

Not in the paper answer:

- Up to **7%** of metastases are <10 HU, so every lesion still needs biochemistry; RCC and hepatoma metastases can also wash out.
- **Unenhanced CT**: **≤10 Hounsfield units (HU)** means a lipid-rich adenoma, which covers >70% of adenomas. Chemical-shift MRI drop-out is about as accurate.

- **CT washout** is the gold standard for lipid-poor lesions, with phases at 0, 60 s and 15 min. It is an adenoma if the absolute percentage washout (APW) is **>60%** or the relative washout (RPW) **>40%**.

ALGORITHM

1. Unenhanced CT: if ≤10 HU or MRI drop-out, then adenoma.
2. If >10 HU, then do a 15-min washout CT.
3. If APW >60% / RPW >40%, then it is an adenoma; if below, manage as indeterminate.

Size	Malignancy (large unselected series)	Surgical series
<4 cm	0.1%	≈2–5% adrenocortical carcinoma (ACC)
4–6 cm	2.4%	≈6–10%
>6 cm	19.5%	≈25%

- **4 cm** is the consensus threshold (sensitive, poorly specific); 6 cm is more specific.
- **Benign imaging overrides size**: no <10 HU lesion was malignant in the largest series.
- Heterogeneity, irregular margins and invasion are suspicious. **Macroscopic fat** (<−30 HU) means myelolipoma. **Calcification is not reassuring**, as it occurs in up to 30% of ACC.
- Classic follow-up re-images at 6 and 12 months; growth of ≥1 cm means consider resection.

Functional imaging

P2 *MIBG scan — isotopes; best in MEN-2; cold in SDHB/VHL*

Not in the paper answer:

- For phaeochromocytoma >5 cm, do MIBG or FDG PET (best in SDHB/VHL) pre-op to exclude metastases. Iodinated contrast does **not** precipitate crisis.

4 · HORMONAL WORK-UP (ALL LESIONS >1 CM)

KEY POINT

Every incidentaloma gets a **1-mg dexamethasone suppression test (DST) and metanephrines**. Add an **aldosterone-to-renin ratio (ARR)** if there is hypertension or unexplained hypokalaemia. **Never biopsy, sample or excise before excluding phaeochromocytoma**.

Incidentalomas comprise cortisol adenoma 5%, phaeochromocytoma 5%, aldosteronoma 1%, ACC 5% and metastasis 2.5%, so **≈18%** are potentially surgical.

Cortisol

P2 *Cushing syndrome and the dexamethasone suppression test — DST cut-offs and pitfalls, MACS*

- 1-mg DST: **≤1.8 µg/dL (50 nmol/L)** excludes autonomy; 1.8–5.0 means possible autonomy; **>5.0 µg/dL (140 nmol/L)** is >95% specific but misses up to 18% of Cushing syndrome.
- Urinary free cortisol (UFC) is insensitive for mild excess and unreliable if GFR is <60 mL/min.

Catecholamines

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Performance	Sensitivity 97–99% – best rule-out	>97% by tandem mass spectrometry
Conditions	**Supine after 20 min rest**, fasting	Normal renal function
Highly indicative	Normetanephrine >2.2 nmol/L and/or metanephrine >1.2 nmol/L (≈4× upper limit of normal)	—

- **False positives** come from paracetamol, mesalazine, sulfasalazine (stop ≥5 days), tricyclics, **phenoxybenzamine**, severe illness and seated sampling.

- If equivocal, repeat supine off β-blockers, check chromogranin A, or do a **clonidine suppression** test.

5 · BIOPSY AND INCIDENTALOMA MANAGEMENT

- Biopsy answers only "is this a metastasis?", because cytology cannot separate adenoma from carcinoma. ESE–ENSAT advises **no biopsy without a history of malignancy**; complications occur in 3–12%, with seeding in ACC.

ALGORITHM

1. If imaging characterises the lesion, then do not biopsy.
2. If it is functional, then do not biopsy.
3. If ACC is suspected, then do not biopsy.
4. If the result will change management (e.g., known primary) and justify risk, then biopsy, **after excluding phaeochromocytoma**.

- Incidentalomas appear on ≈5% of scans; ≈93% are inactive and need no treatment.

6 · PRIMARY ALDOSTERONISM (PA)

P2 *How will you evaluate and manage 38 yr male with uncontrolled hypertension — ARR, saline, CT, AVS; cure ≈94%*

Not in the paper answer:

- PA causes 3–13% of hypertension; most patients (63–91%) are **normokalaemic**.

Subtype	%	Surgically curable?
Bilateral (idiopathic) hyperplasia	60	No – mineralocorticoid receptor antagonist (MRA)
Aldosterone-producing adenoma	35	Yes
Unilateral hyperplasia	2	Yes
Carcinoma, ectopic (ovarian), familial hyperaldosteronism (FH)	<1 each	Carcinoma/ectopic yes; FH no

- **FH-I** (glucocorticoid-remediable) results from a chimeric gene that puts aldosterone under ACTH control; it causes early severe hypertension and is treated with glucocorticoid.
- **Screen (Endocrine Society 2016)**: resistant hypertension (≥3 drugs including a diuretic) or control on ≥4 drugs; BP ≥150/100 on three days; hypertension and hypokalaemia; **incidentaloma with hypertension**; sleep apnoea; onset <20 y or stroke <50 y; family history of PA; disproportionate target-organ damage.
- **ARR** is measured in the morning, seated, as plasma aldosterone concentration (PAC, ng/dL) divided by plasma renin activity (PRA, ng/mL/h). It is positive if **ARR >20–30 with PAC >15–20 ng/dL**. Floor PRA at 0.2 and correct K^+ first.

Drug	Effect on ARR	Action
Spironolactone, eplerenone, amiloride	Profound	Stop ≥4–6 weeks; use α1-blocker or calcium channel blocker (CCB)
β-blockers	Lower renin, so **false positive**	May continue (or stop)
ACE inhibitors / angiotensin receptor blockers	Raise renin, so false negative	Continue; undetectable renin is suspicious
Calcium channel blockers	Neutral	Continue

Confirmatory test	Protocol	Positive
Oral sodium loading	3 days high salt; 24-h urine Na >200 mmol	Urine aldosterone >12 µg/day

Confirmatory test	Protocol	Positive
IV saline infusion	2 L 0.9% saline over 4 h, recumbent	PAC >5 ng/dL (>10 suggests adenoma)
Captopril challenge	25–50 mg, when salt load is contraindicated	PAC stays >15 ng/dL

- **Skip confirmation** if there is spontaneous hypokalaemia, undetectable renin and PAC >20 ng/dL.
- CT all surgical candidates to exclude ACC, but ≈20% of aldosteronomas are <1 cm, so CT alone misdirects ≈25%.
- **AVS** (sensitivity 95%, specificity 100% if done well) is done in the morning with K^+ corrected, sampling both adrenal veins and the IVC for aldosterone and cortisol.
- **Selectivity** proves the catheter is in the vein (adrenal-to-IVC cortisol >1.1–5:1). **Lateralisation**: a cortisol-corrected aldosterone ratio **>2–4:1** indicates ipsilateral adrenalectomy.
- **Omit AVS** if age <40 with a clear unilateral adenoma >1 cm and a normal contralateral gland, or if ACC is suspected.

ALGORITHM

1. If there is a screening indication, measure the ARR.
2. If the ARR is positive, confirm with a suppression test. If there is a family history or onset <20 y, test for FH-I.
3. If not a surgical candidate, then MRA.
4. If a surgical candidate, do CT. If age <40 with a unilateral >1 cm nodule, then adrenalectomy.
5. Otherwise do AVS: if lateralised, laparoscopic adrenalectomy; if not lateralised, MRA.

- After surgery, check PAC and ARR; † stop K^+ and MRA, and watch for **hyperkalaemia** for weeks because the contralateral glomerulosa is suppressed. BP settles over 6 months; residual hypertension is likelier in older, obese men with long-standing disease.
- MRA: spironolactone 25–50 mg, up to a maximum of 400 mg (gynaecomastia, erectile dysfunction); eplerenone 25 to 100 mg (selective).

7 · CUSHING SYNDROME AND CORTISOL EXCESS

P2 *Cushing syndrome and the dexamethasone suppression test — causes, IPSS, Nelson, pituitary cure*

Not in the paper answer:

- Post-op insufficiency follows ≈65% of MACS resections; the HPA axis recovers in a median of 6.5 (MACS) to 11 (Cushing) months.
- **Urological**: hypogonadotropic hypogonadism and **urolithiasis in up to 50%**, so test cushingoid stone formers.

ALGORITHM

1. Exclude exogenous steroid.
2. Screen with 1-mg DST, late-night salivary cortisol (LNSC) ×2 or UFC ×2.
3. If abnormal, endocrinologist confirms; measure **morning ACTH**.
4. If ACTH low: adrenal, do CT.
5. If normal/high: pituitary MRI and **inferior petrosal sinus sampling after CRH**. A central-to-peripheral gradient means Cushing disease; no gradient means ectopic.

- **Bilateral adrenalectomy** is for failed pituitary treatment, life-threatening hypercortisolism, or unresectable or occult ectopic ACTH. The cost is lifelong replacement, crisis risk and **Nelson syndrome** (chiasmal compression, oculomotor palsies).

	ACTH-independent macronodular adrenal hyperplasia (AIMAH)	Primary pigmented nodular adrenocortical disease (PPNAD)
Glands	Massive, nodules up to 4 cm, >60 g each	Normal size; small black/brown nodules in atrophic cortex

	ACTH-independent macronodular adrenal hyperplasia (AIMAH)	Primary pigmented nodular adrenocortical disease (PPNAD)
Association	McCune–Albright	Carney complex (≈half)
Clue	Looks ACTH-driven – check ACTH	Paradoxical rise in UFC after dexamethasone
Treatment	Bilateral adrenalectomy; unilateral removal of the larger gland has worked in AIMAH	

- **Mild autonomous cortisol secretion (MACS)**: individualise surgery for hypertension, obesity, dysglycaemia or osteopenia; progression to overt Cushing is very unlikely.
- **Post-op adrenal insufficiency is expected** after overt Cushing (≈99.7%), because the contralateral gland is atrophic.
- Give hydrocortisone 100 mg IV on the day of surgery, then an oral taper (25–50 mg/day). Alternatively give dexamethasone 0.5 mg 6-hourly, where a morning cortisol <50 nmol/L confirms cure. Maintenance is 10–12 mg/m²/day in 2–3 doses.

8 · PHAEOCHROMOCYTOMA AND PARAGANGLIOMA (PPGL)

These can kill on the table, so block before surgery.

P2 *Phaeochromocytoma: overview, pre-operative and intra-operative management — genetics, blockade doses, targets, post-op care, lifelong metanephrines*

Not in the paper answer:

- ≈24% of apparently sporadic cases carry germline mutations.
- 75–90% are adrenal; paraganglioma is extra-adrenal. Headache occurs in 60–90%.

> **KEY POINT**
> The rule of 10s is outdated – every patient with PPGL should be offered genetic counselling/testing (Endocrine Society 2014).

Syndrome	Gene	Features	PPGL risk / malignancy
MEN-2A	RET 10q11.2	Medullary thyroid cancer (MTC), hyperparathyroidism	50% / ≈3%; adrenal, bilateral
MEN-2B	RET	MTC, mucosal neuromas, marfanoid	50% / ≈3%
VHL type 2	VHL 3p25	RCC, haemangioblastoma, epididymal cystadenoma	10–20% / ≈5%; normetanephrine only
Neurofibromatosis 1 (NF1)	NF1 17q11	Neurofibromas, café-au-lait	1% / ≈11%
PGL-4	**SDHB** 1p36	Abdominal/thoracic paragangliomas	≈20% / **30–50%**
PGL-1	SDHD 11q23 (**maternal imprinting** – paternally transmitted)	Head and neck, multifocal	≈20% / <3%

All are autosomal dominant. VHL type 1 has no phaeochromocytoma, whereas type 2 does.

- On CT the mean is ≈35 HU; the T2 "light bulb" sign is unreliable.
- **Block all with abnormal biochemistry, even if normotensive.** Phenoxybenzamine causes hypotension after resection; α1-selective drugs do so less. CCBs can be sole therapy in mild cases.
- **Metyrosine** 250 mg 2–3 times a day, increased to 1.5–2 g, is added to α-blockade for large or metastatic tumours; it causes sedation and **extrapyramidal** effects.

KEY POINT

Why never β before α? β-blockade removes β2-mediated arteriolar dilatation. Catecholamines then act unopposed on α1 receptors and cause a hypertensive crisis. Hence use selective β1 agents.

- **Targets**: seated BP 120–130/80; standing systolic >80–90 mmHg; † heart rate 60–70/min. **Volume repletion** is key.
- In **MEN-2 and VHL**, do **cortical-sparing partial adrenalectomy**.
- Malignancy is defined **only by metastases**; risk factors are SDHB, extra-adrenal site, >5 cm and dopamine secretion.
- Check metanephrines at ≈6 months, then annually for life, because recurrence occurs even >15 years later.

9 · NON-FUNCTIONAL BENIGN LESIONS

Lesion	Key features	Management
Myelolipoma	Fat and haematopoietic tissue; macroscopic fat diagnostic; may exceed 10 cm	Exception to 4-cm rule; still do hormonal work-up. Excise if symptomatic or large; embolise a bleed
Cysts	Endothelial 45%, pseudocyst 39% (harbours malignancy ≈7%); >30 HU means haemorrhage	Endocrine work-up. **Excise if >5 cm, thick-walled, heterogeneous or symptomatic**
Ganglioneuroma	Young; encases vessels without narrowing them	Benign; diagnosed at resection
Oncocytoma	Rare; large; ≈10% functional	Resect; >30% malignant (unlike renal)

10 · ADRENOCORTICAL CARCINOMA (ACC)

P4 *Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management — syndromes, CT, Weiss, ENSAT, R0 surgery, mitotane, trials, surveillance*

Not in the paper answer:

- Overall 5-year survival is 20–47% (stage IV <5%), and 60–80% recur after resection.

ENSAT stage	Definition	At diagnosis
I	T1 (≤5 cm) N0 M0	3–4%
II	T2 (>5 cm) N0 M0	29–46%
III	T3 (periadrenal fat) or T4 (adjacent organ) or N1 or venous thrombus	11–19%
IV	Distant metastasis only	39–49%

- Resect recurrences if low grade with a long disease-free interval.
- † Mitotane often needs higher glucocorticoid doses, because it raises binding globulins and speeds clearance.
- **Paediatric ACC** is >90% functional (virilisation), warrants TP53 testing and has a better prognosis.

11 · ADRENAL METASTASES

- With known malignancy, **>50%** of new adrenal lesions are metastases. RCC and hepatoma deposits may wash out, so growth and biopsy decide. **Still exclude phaeochromocytoma**.
- Resect an isolated metastasis in selected patients: non-small cell lung cancer (level 1 evidence) and melanoma, and RCC as part of cytoreduction.

12 · ADRENAL INSUFFICIENCY

	Primary (Addison)	Secondary (pituitary/hypothalamic)
Causes	Autoimmune (anti-21-hydroxylase), **tuberculosis**, bilateral haemorrhage, infiltration, **bilateral adrenalectomy**, drugs (ketoconazole, abiraterone)	**Exogenous glucocorticoid suppression** (commonest for surgeons), pituitary tumour/apoplexy, Sheehan, critical illness
Aldosterone	Lost	Preserved
Skin	Hyperpigmentation (high ACTH)	Pallor
Electrolytes	Low Na^+, **high K^+**; salt craving	Low Na^+, normal K^+
Replacement	Hydrocortisone + fludrocortisone	Hydrocortisone only

- A **crisis** mimics an acute abdomen. Those at risk have a cortisol-secreting tumour, bilateral adrenalectomy or a **previous contralateral nephrectomy/adrenalectomy**, so review old notes.
- Diagnose with morning cortisol and ACTH, then ACTH stimulation. Give ½–⅔ of the hydrocortisone dose on waking.

KEY POINT

† **Adrenal crisis**: do not wait for results. Draw cortisol and ACTH, give **hydrocortisone 100 mg IV bolus**, then 200 mg/24 h (infusion or 50 mg 6-hourly). Give 0.9% saline (1 L in the first hour) with dextrose, treat the precipitant, and taper over 1–3 days. Fludrocortisone is unnecessary while hydrocortisone is >50 mg/day. Teach sick-day rules and issue a steroid card.

Surgical stress (chronic steroids)	Example	Cover
Minor	Inguinal hernia, <1 h	Hydrocortisone 25 mg
Moderate	Joint replacement	Usual dose + hydrocortisone 50–75 mg intra-op
Major	Oesophagectomy, bypass	Usual dose + hydrocortisone 50 mg IV 8-hourly for 48–72 h

Glucocorticoid	Potency	Replacement (mg/day)	Half-life (h)
Hydrocortisone	1	20	8–12
Prednisone / prednisolone	4	5	18–36
Dexamethasone	25–50	0.5	36–54

13 · CONGENITAL ADRENAL HYPERPLASIA (CAH)

- CAH is autosomal recessive, and **21-hydroxylase deficiency is >95%**; low cortisol raises ACTH, causing hyperplasia and androgen excess (see Ch 59).
- Non-classic CAH may present as an adult mass, and testicular rests mimic tumour, so consider them before orchidectomy.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SPARTACUS	RCT: CT- vs AVS-guided PA treatment	Same antihypertensive intensity at 1 y; CT and AVS discordant in half; persistent aldosteronism 20% (CT) vs 10% (AVS)	AVS remains standard for biochemical cure
FIRM-ACT	RCT: mitotane + etoposide, doxorubicin, cisplatin (EDP) vs mitotane + streptozotocin, advanced ACC	Better response and progression-free survival; overall survival similar	M-EDP first-line

Trial	Design	Result	Take-home
ADJUVO	RCT: adjuvant mitotane vs observation, low/intermediate-risk R0 stage I–III ACC	No difference in recurrence	No routine adjuvant mitotane in low risk

15 · MCQ ONE-LINERS

MCQ ONE-LINERS

1. Commonest ectopic ACTH source: bronchial carcinoid.

147 Surgery of the Adrenal Glands

P2 *Adrenal Sx. — indications, contraindications, preparation, approaches, steps, complications*

Not in the paper answer:

- Size and imaging criteria for incidentaloma (§2).

1 · SURGICAL ANATOMY (ESSENTIALS)

- The **right** gland lies behind the bare area of the liver. Its vein is **<1 cm** long and enters the **posterior IVC at 45°**. The **left** gland is more mobile.

2 · INDICATIONS

Two gates: any functional tumour, or malignant potential.

Category	Indication
Functional (any size)	Aldosteronoma; cortisol adenoma; Cushing disease after failed trans-sphenoidal surgery; phaeochromocytoma (PCC); virilising/feminising tumour
Non-functional	Adrenocortical carcinoma (ACC); symptomatic cyst/myelolipoma; solitary metastasis
Size	**≥4 cm** ; >2 cm growing ≥1 cm in 12 months
Imaging	Non-contrast CT **>10 HU**; washout <50% with delayed >35 HU; chemical-shift MRI not adenoma; FDG-PET uptake equal to or above liver

KEY POINT

Trap: bilateral hyperplasia causing aldosteronism is **not** surgical, so adrenal venous sampling (AVS) comes before adrenalectomy for aldosteronism.

Incidentaloma work-up

- ACC is found in ~2% and autonomous secretion in up to 20%. Measure sex steroids if ≥4 cm or virilised.
- All get a 1-mg dexamethasone suppression test (never spot cortisol) and metanephrines (>2× upper limit is significant).
- If hypertensive, check the aldosterone-to-renin ratio (ARR >20 with aldosterone >20 ng/dL), then salt loading, then AVS; up to 40% are normokalaemic.
- Inert, ≤4 cm, ≤10 HU lesions get CT at 12 months. Operate on myelolipoma (≤ −20 HU) only if **>7 cm**, symptomatic or atypical.

3 · CONTRAINDICATIONS

- Correct physiology first. Low-dose aspirin may continue; necessary surgery in pregnancy can be done in any trimester. Confirm that a prior nephrectomy has not removed the ipsilateral adrenal before contralateral adrenalectomy.

4 · PRE-OPERATIVE PREPARATION

Pathology	Preparation	Why
PCC	α-blockade for **2 weeks** (phenoxybenzamine; doxazosin, calcium channel blocker, metyrosine alternatives); β **only after α**; volume expansion	Contracted patients collapse once vein tied
Cushing	†Hydrocortisone 100 mg at induction, tapered over months; thromboprophylaxis	Contralateral gland is suppressed
Aldosteronoma	†Correct K^+; control BP	†Hypokalaemia causes arrhythmia

5 · APPROACHES

- **Minimally invasive surgery is standard** (since 1992).
- The **retroperitoneoscopic** route (lateral or **prone**) avoids bowel, so it suits prior abdominal surgery; bilateral small tumours can be done prone without repositioning.
- **Open** surgery is for suspected **ACC**, large tumours, or invasion of the IVC or organs.

6 · OPERATIVE STEPS

Control the vein, then free the gland.

ALGORITHM

1. **Right**:
2. Elevate the liver; Kocher the duodenum; expose the IVC.
3. Dissect up from the renal vein to the short adrenal vein.
4. **Warn anaesthesia**, then clip early.
5. Dissect superiorly and laterally down to psoas; bag the gland intact.

ALGORITHM

1. **Left**:
2. Reflect the colon at Toldt, outside Gerota; lift spleen and pancreas.
3. Follow the left renal vein to the adrenal vein; clip after warning.
4. Protect pancreas and splenic artery; divide superomedial tissue; bag.

- **Retroperitoneoscopic**: enter below the 12th rib tip with a balloon. Along psoas the **renal artery is met first**; the right vein enters the posterolateral IVC above the hilum.

KEY POINT

Ligate the vein early in PCC to limit the catecholamine surge; hypotension follows, so tell anaesthesia first.

7 · SPECIAL PATHOLOGIES

- **PCC intra-op**: treat a surge with nitroprusside, nitroglycerin or phentolamine, and arrhythmia with esmolol.
- **ACC** needs a high-volume surgeon (>6, ideally ≥20/year) and an **open, intact, en bloc** resection.
- **Metastasis** (two-thirds from lung or kidney) is resected when the adrenal is the only disease; renal origin and <5 cm do best.
- Avoid ablation of PCC or ACC outside trials.

8 · PARTIAL (CORTICAL-SPARING) ADRENALECTOMY

- **Indications**: bilateral or hereditary PCC, familial syndromes, prior contralateral adrenalectomy, children, Conn adenoma and cysts.

- Arteries enter superiorly, medially and inferiorly, so **lateral dissection is safe**; the main vein may be divided. †Retain about one-third of a gland, guided by intra-op ultrasound.

9 · COMPLICATIONS

- For bleeding, apply pressure, raise the pneumoperitoneum, side-bite the IVC, and convert if uncontrolled.
- †Check glucose hourly for 24–48 h after PCC. Treat insufficiency with IV hydrocortisone 100 mg and saline. For hyperkalaemia after aldosteronoma, stop K^+ and spironolactone, occasionally adding fludrocortisone.

10 · OUTCOMES

- PCC recurs in up to **16%** over life, so check BP and metanephrines annually.
- †Aldosteronoma surgery corrects hypokalaemia in nearly all and cures hypertension in about half. Advanced-stage ACC has a 5-year survival **<15%**.

140 Retroperitoneal Tumours

1 · ANATOMY OF THE RETROPERITONEUM

P1 *Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. — boundaries, contents, spaces, vessels, nerves at risk*

Not in the paper answer:

- **Depth of psoas invasion predicts nerve injury**: genitofemoral injury causes thigh paraesthesia, while femoral injury weakens hip flexion and knee extension.

> **KEY POINT**
> The operative problem is usually the neighbouring structure, not the tumour: RP tumours grow silently (resected sarcomas average **>15 cm**) and present with multi-organ involvement.

2 · CLASSIFICATION

The group decides first treatment.

Group	Entities	First treatment
Germ cell / gonadal	Seminoma, nonseminoma, sex cord stromal	Platinum chemotherapy (germ cell tumour, GCT); RP lymph node dissection, RPLND (sex cord stromal)
Mesodermal — malignant	Liposarcoma, leiomyosarcoma, undifferentiated pleomorphic sarcoma (UPS), synovial, Ewing, desmoplastic small round cell tumour (DSRCT), unclassified	Surgery; chemotherapy (Ewing, DSRCT)
Mesodermal — benign/intermediate	Lipoma, leiomyoma, solitary fibrous tumour (SFT), PEComa, desmoid, myelolipoma, GIST	Excision / observation
Neurogenic (~10%)	Schwannoma, neurofibroma, malignant peripheral nerve sheath tumour (MPNST), ganglioneuroma, neuroblastoma, paraganglioma	Surgery
Haematological (~1/3 of primary RP)	Lymphoma, plasmacytoma	Chemotherapy — biopsy only
Cystic	Lymphangioma, mucinous cystadenoma, cystic mesothelioma, cystic teratoma	Complete excision

Group	Entities	First treatment
Unknown primary	Poorly differentiated carcinoma encasing vessels	Histology-led

3 · CLINICAL EVALUATION

Tissue only if it changes treatment.

- Most are painless or incidental. The sequence is **imaging, then serology, then tissue**.
- **CT chest/abdomen/pelvis** comes first (NCCN). **Negative HU (fat)** suggests liposarcoma, angiomyolipoma or myelolipoma, but CT cannot tell origin or malignancy. **MRI** shows vascular involvement; PET-CT is selective.
- In a young man check **AFP and hCG**; in an older man, PSA. Add blood count and electrophoresis; † metanephrines if paraganglioma is possible.
- **Biopsy** only when it **changes first treatment**: lymphoma, GCT, Ewing, or high-grade sarcoma for neoadjuvant therapy. Use a **core (14–16 G coaxial) rather than FNA**; † take a retroperitoneal route, with the tract inside the future resection field.
- Avoid biopsy of purely cystic lesions, † an unblocked paraganglioma, or a young man with raised AFP/hCG.

4 · MESENCHYMAL TUMOURS (SARCOMA)

- Soft-tissue sarcoma is <1% of adult cancers; **15–20%** arise in the RP.

Entity	Key features	Treatment
Well-/de-differentiated (WD/DD) liposarcoma (commonest, **~60%**)	Fat density; **MDM2/CDK4 (12q) amplification**, ring chromosomes; S100+	Radical compartmental resection
Leiomyosarcoma (**~20%**)	From vessel media (IVC); no fat; SMA, desmin, h-caldesmon+	Negative margins + adjuvant
UPS (ex-MFH)	5th–6th decade; no fat, peripheral nodular enhancement	Wide resection
Unclassified (≤20%)	Exclusion; prior RT in a quarter	Screen MDM2
Synovial	Calcification; TLE1; **t(X;18)**	Chemosensitive
SFT	**NAB2-STAT6**; Doege-Potter hypoglycaemia (IGF)	En bloc; radiosensitive
Ewing	EWSR1-FLI1 t(11;22); infiltrative	Chemotherapy-based
DSRCT	Young men; peritoneal implants; WT1-EWSR1 t(11;22)	Multimodal; poor
PEComa	HMB-45 + muscle markers; TSC2	Excision; mTOR inhibitors if unresectable
Desmoid	Nuclear β-catenin; APC/CTNNB1; never metastasises; recurs in up to half	Surgery or observation

- **Liposarcoma**: **dense nodules in a fatty mass mean de-differentiation** and are the biopsy target. WD recurs locally in up to 40% but rarely metastasises; DD spreads distantly in up to 20%. Mean time to recurrence is ~**2 years**, so surveillance is lifelong.
- If unresectable, anthracycline or trabectedin offers modest benefit. **Pleomorphic liposarcoma** is MDM2-negative and metastasises early to lung.
- **Leiomyosarcoma** obstructs veins, urine and bowel; biopsy underestimates its aggressiveness.

KEY POINT

MDM2 amplified (WD/DD) means local disease, won or lost at the margin; MDM2 absent (pleomorphic) means systemic disease. Infiltrative types (Ewing, DSRCT) need chemotherapy because no margin is achievable.

- † **FNCLCC grade** adds differentiation (1–3), mitoses per 10 high-power fields (1: 0–9, 2: 10–19, 3: ≥20) and necrosis (0 none, 1 <50%, 2 ≥50%): **2–3 = G1, 4–5 = G2, 6–8 = G3**. Grade and histology, not size, predict distant spread.

5 · NEUROGENIC TUMOURS

Tumour	Key features	Management
Schwannoma	Paravertebral; encapsulated; S100+; slow	Local resection curative (often enucleable)
Neurofibroma	NF-1 or sporadic; malignant change up to 20%	Surgical
MPNST	<10% of RP sarcoma; half NF-1; H3K27me3 loss; spreads along roots	Resect with **parent nerve**; 5-year ~70% R0 vs ~20% R1
Ganglioneuroma	3rd–4th decade; calcification up to 30%	Excision
Neuroblastoma	MYCN adverse; MIBG staging	Paediatric chemotherapy with or without surgery
Paraganglioma	Commonest at organ of Zuckerkandl (below IMA); T2-bright; secretory ~40%; metastases 20–50%; NF-1, MEN-2A, SDHB	Surgery after α-blockade† (see Ch 145)

6 · GERM CELL, HAEMATOLOGICAL, CYSTIC, UNKNOWN PRIMARY

- **GCT**: an RP mass in a young man is GCT until proven otherwise; extragonadal disease is often a **burnt-out testicular primary**. Give platinum first, then resect all nonseminomatous residual masses, because teratoma is chemoresistant and marker-silent (see Ch 79–80).
- **Sex cord stromal** tumours resist chemo and radiotherapy, so do early RPLND in high-risk node-negative primaries.
- **Lymphoma** forms confluent nodal masses: biopsy, never resect.
- **Cystic** lesions (lymphangioma, mucinous cystadenoma, cystic mesothelioma) need **complete excision**, because aspiration or marsupialisation recur.
- **Unknown primary**: if midline or i(12p)-positive, give platinum; a single deposit is excised or irradiated.

KEY POINT

Histology decides first treatment: exclude chemotherapy-first entities (GCT, lymphoma, Ewing, midline unknown primary) before planning sarcoma resection.

7 · SURGERY FOR RP SARCOMA

The first operation decides cure.

- **Complete en bloc (R0) resection at the first operation** in an experienced centre is the best chance.
- **Compartmental resection** removes adherent uninvolved neighbours (kidney, colon, psoas fascia), which lowers local recurrence. Preserve duodenum, pancreatic head, major vessels and femoral nerve unless invaded. † **Never enucleate**, because the pseudocapsule contains tumour.
- Nephrectomy is needed in up to **55%** but rarely leads to dialysis; † get a split renogram first.
- After **bleomycin**, keep FiO_2 low and fluids restrictive.

- Through a midline laparotomy, **split and roll** over the great vessels and ligate lumbar vessels and retrocrural lymphatics to prevent chylous ascites.
- The **aorta is always reconstructed**. A chronically occluded **IVC** tolerates ligation; IVC grafts thrombose in 10–40%, so anticoagulate for ≥3 months.

8 · RPLND IN GERM CELL TUMOURS

P1 *Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — landing zones, templates, PC-RPLND, nerve-sparing*

- Primary RPLND gives survival ~100% with either template. In post-chemotherapy RPLND, a full bilateral infrahilar template is standard; unilateral only if the mass is **≤5 cm** in the landing zone, markers are normal, and IGCCCG risk is good or intermediate.
- Sparing the L1–L4 fibres and the superior hypogastric plexus gives antegrade ejaculation in **90–100%**.

9 · COMPLICATIONS

P1 *Lymphocele and postoperative lymphatic leak (chylous ascites) — risk, diagnosis, stepwise treatment*

Not in the paper answer:

- Chylous ascites occurs in 2–7% after chemotherapy. Other risks are pancreatic leak (check drain amylase), VTE after caval reconstruction and femoral neuropraxia.

ALGORITHM

1. **Chylous ascites**:
2. Paracentesis and a medium-chain triglyceride (MCT) diet, with or without octreotide.
3. If it reaccumulates, place a drain.
4. Then nil by mouth and total parenteral nutrition.
5. If output stays >100 mL/day, then peritoneovenous (LeVeen) shunt or ligation.
6. Continue the MCT diet for 1–3 months after resolution.

10 · ADJUVANT AND NEOADJUVANT THERAPY

Neither is routine; surgery is the mainstay.

- **Radiotherapy**: pre-op is better tolerated than post-op. **STRASS** showed no gain, so it is **not standard**; † an exploratory benefit in low-grade liposarcoma keeps it an option.
- **Chemotherapy**: WD liposarcoma and leiomyosarcoma are largely resistant; myxoid liposarcoma, synovial sarcoma and Ewing respond. **Standard neoadjuvant chemotherapy is only for Ewing**.

11 · RECURRENCE AND FOLLOW-UP

- † Local recurrence kills in liposarcoma, whereas leiomyosarcoma fails distantly (lung, liver).
- † CT every 3–6 months for 2–3 years, 6-monthly to 5 years, then annually; lifelong for liposarcoma.
- Re-resect slow-growing resectable recurrence (WD liposarcoma).

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
EORTC-62092 STRASS	Phase III, 266 patients: pre-op RT + surgery vs surgery	No abdominal recurrence-free survival difference (HR 1.01); more complications	Pre-op RT not standard
STRASS II (ongoing)	Neoadjuvant doxorubicin + ifosfamide/dacarbazine + surgery vs surgery	Awaited	Neoadjuvant chemotherapy unproven

★ Putting it together: adrenal and retroperitoneal decisions

A · ADRENAL INCIDENTALOMA: TWO QUESTIONS

ALGORITHM

1. **Malignant?** If unenhanced CT is ≤10 HU or MRI shows drop-out, then adenoma. If >10 HU, do washout CT (APW >60% / RPW >40% = adenoma). If >4 cm, suspicious or growing, consider resection.
2. **Functioning?** Test all with 1-mg DST and metanephrines, plus ARR if hypertensive or hypokalaemic.
3. If **silent, <4 cm and benign**, then no follow-up.

B · FUNCTIONAL TUMOURS COMPARED

	Aldosteronism	Cushing	Phaeo
Screen	ARR	1-mg DST, salivary or urinary cortisol	Metanephrines
Next	Confirm, then CT, then AVS	ACTH: low = adrenal; normal/high = pituitary/ectopic	CT/MRI, functional imaging; genetics
Pre-op	Correct K^+	Steroid cover	α then β; fluids, salt
Post-op	Stop K^+/spironolactone; hyperkalaemia	Steroids until axis recovers	Hypotension, hypoglycaemia
Medical	Mineralocorticoid antagonist (bilateral)	Steroidogenesis inhibitors	Long-term α-blockade

C · ADRENALECTOMY APPROACH

KEY POINT

Use laparoscopy for benign tumours. Operate open for large suspected carcinoma or invasion, because rupture causes recurrence. On the right, control the short caval vein carefully. Partial adrenalectomy suits bilateral or hereditary disease.

D · RETROPERITONEAL MASS

ALGORITHM

1. **Image**, then **exclude** paraganglioma (metanephrines), germ cell tumour (testis, markers) and lymphoma.
2. If it changes treatment, do a **core biopsy** by the retroperitoneal route.
3. **Sarcoma:** en bloc R0 resection at the first operation in a specialist centre; never enucleate. Follow liposarcoma with lifelong CT.